Software Engineering Lecture 1: Agile Methodologies

Sprint Iterations and User Story Estimation

Agile software development values rapid iterative delivery of working software over exhaustive upfront documentatio
Scrum teams plan work in time-boxed sprint iterations, tracking velocity through story points and burndown charts.
User stories capture business value from the persona perspective using standard acceptance criteria.

Software Engineering Lecture 2: Architectural Patterns

SOLID Design Principles and Clean Architecture

Clean architecture separates business domain logic from volatile database drivers and web presentation frameworks
The Single Responsibility Principle asserts that a class or module should have only one reason to change.
Dependency inversion dictates that high-level business policies should depend on abstractions rather than low-level

Software Engineering Lecture 3: Continuous Integration and Testing

Automated Testing Pyramids and Deployment Pipelines

Continuous integration mandates that developers merge code changes frequently into a shared repository with auto
The testing pyramid prioritizes fast isolated unit tests at the base, followed by integration tests and end-to-end tests.
Automated canary deployment strategies gradually shift user traffic to new revisions while observing health telemetr